# EE2FH4
# Electromagnetics I

**Term II, January – April 2026**

# MATLAB Examples and Exercises (Set 4)

**Prepared by: Dr. M. H. Bakr and C. He**

**Example:** A surface charge of 5.0 $\mu$C/m$^2$ is located in the $x$-$z$ plane in the region -2.0 m $\leq x \leq$ 2.0 m and -3.0 m $\leq y \leq$ 3.0 m. Find analytically the electric field at the point (0, 4.0, 0) m. Verify your answer using a MATLAB program that applies the principle of superposition.

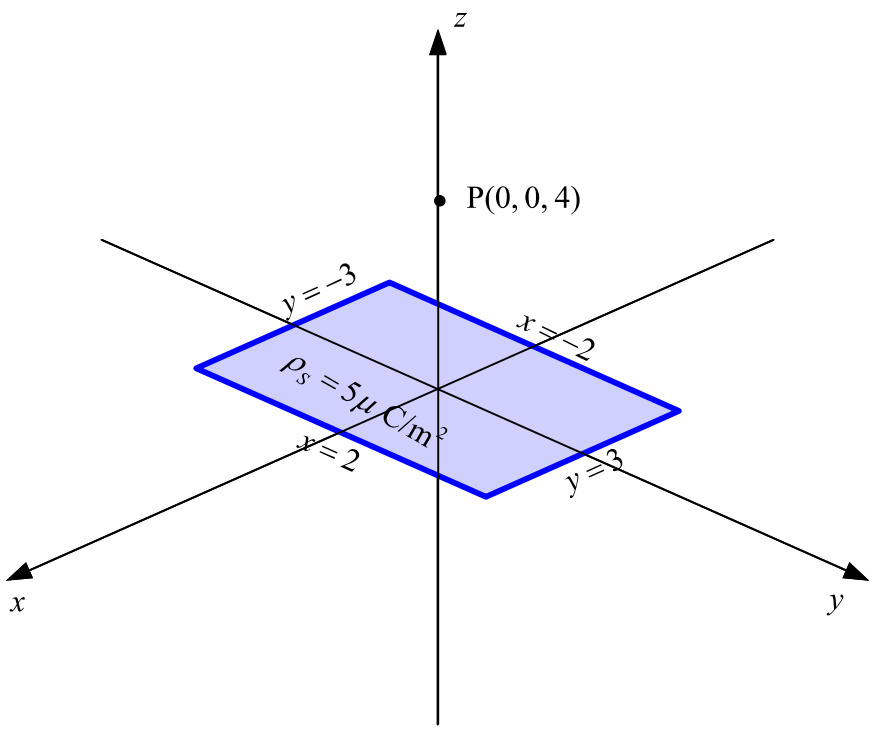


**Figure 4.1 The surface charge of the example of Set 4.**

**Analytical solution:**
As can be seen in Figure 4.2 , for any point A on the surface charge, we can find another point A' whose electric field at P has the same magnitude but the opposite sign of that of A in the direction parallel to the surface charge. Hence the electric field at P has only a $z$ component .

$$dQ = \rho_s dS = \rho_s dxdy$$

$$dE = \frac{dQ}{4\pi\varepsilon |\mathbf{R}|^2}$$

$$dE_z = dE\cos\theta$$

$$\cos\theta = \frac{|\mathbf{R}_z|}{|\mathbf{R}|} = \frac{4}{\sqrt{(x-0)^2+(y-0)^2+(4-0)^2}} = \frac{4}{\sqrt{x^2+y^2+16}}$$

$$dE_z = \frac{\rho_S dxdz}{4\pi\varepsilon|\mathbf{R}|^2}\frac{4}{\sqrt{x^2+y^2+16}} = \frac{\rho_S}{\pi\varepsilon\left(\sqrt{x^2+z^2+16}\right)^3}dxdy$$

$$E_z = \iint_s dE_z$$

$$= \int_{y=-3}^{y=3}\int_{x=-2}^{x=2}\frac{\rho_S}{\pi\varepsilon\left(\sqrt{x^2+y^2+16}\right)^3}dxdy$$

$$= \frac{\rho_S}{\pi\varepsilon}\int_{y=-3}^{y=3}\int_{x=-2}^{x=2}\frac{1}{\left(\sqrt{x^2+y^2+16}\right)^3}dxdy$$

$$= \frac{\rho_S}{\pi\varepsilon}\int_{y=-3}^{y=3}\int_{x=-2}^{x=2}\frac{1}{\left(\sqrt{x^2+a^2}\right)^3}dxdy \quad (\text{let } a^2 = y^2+16)$$

$$= \frac{\rho_S}{\pi\varepsilon}\int_{y=-3}^{y=3}\left.\frac{x}{a^2\sqrt{a^2+x^2}}\right|_{x=-2}^{x=2}dy$$

$$= \frac{\rho_S}{\pi\varepsilon}\int_{y=-3}^{y=3}\frac{4}{\left(y^2+16\right)\sqrt{y^2+20}}dy$$

let $y = \sqrt{20}\tan\alpha$

then $dy = \sqrt{20}\sec^2\alpha d\alpha,\ \sqrt{y^2+20} = \sqrt{20}\sec\alpha,$

and $y^2+16 = 20\tan^2\alpha + 16$

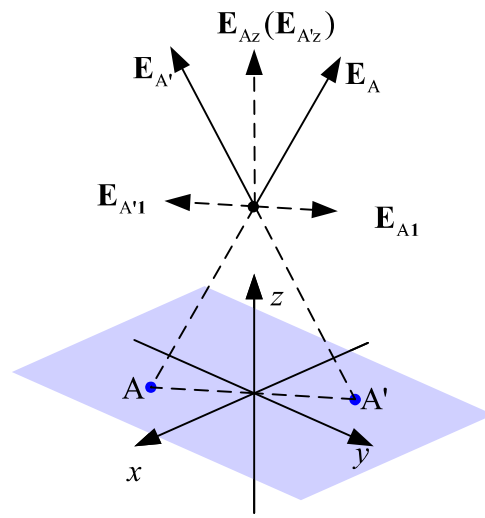


**Figure 4.2 The field components.**

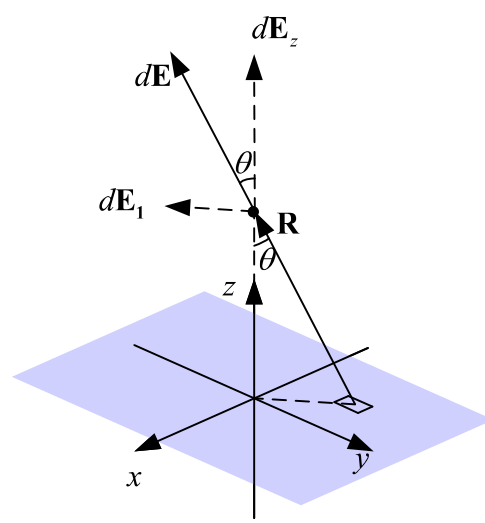


**Figure 4.3 Field decomposition.**

therefore

$$E_z = \frac{4\rho_S}{\pi\varepsilon}\int_{\alpha=\alpha_1}^{\alpha=\alpha_2} \frac{\sqrt{20}\sec^2\alpha\, d\alpha}{\left(\sqrt{20}\sec\alpha\right)\left(20\tan^2\alpha+16\right)}$$

$$= \frac{4\rho_S}{\pi\varepsilon}\int_{\alpha=\alpha_1}^{\alpha=\alpha_2} \frac{\sec\alpha\, d\alpha}{20\tan^2\alpha+16}$$

$$= \frac{4\rho_S}{\pi\varepsilon}\int_{\alpha=\alpha_1}^{\alpha=\alpha_2} \frac{\frac{1}{\cos\alpha}d\alpha}{20\frac{\sin^2\alpha}{\cos^2\alpha}+16}$$

$$= \frac{4\rho_S}{\pi\varepsilon}\int_{\alpha=\alpha_1}^{\alpha=\alpha_2} \frac{\cos\alpha\, d\alpha}{20\sin^2\alpha+16\cos^2\alpha}$$

$$= \frac{4\rho_S}{\pi\varepsilon}\int_{\alpha=\alpha_1}^{\alpha=\alpha_2} \frac{\cos\alpha\, d\alpha}{4\sin^2\alpha+16} = \frac{\rho_S}{\pi\varepsilon}\int_{y=-3}^{y=3} \frac{\cos\alpha\, d\alpha}{\sin^2\alpha+4}$$

let $u = \sin\alpha$ then $du = \cos\alpha\, d\alpha$ therefore

$$E_y = \frac{\rho_S}{\pi\varepsilon}\int_{u=u_1}^{u=u_2} \frac{du}{u^2+4}$$

$$= \frac{\rho_S}{\pi\varepsilon}\times\frac{1}{2}\arctan\frac{u}{2}\Bigg|_{y=-3}^{y=3}.$$

as we can see from Figure 4.4, $y = \sqrt{20}\tan\alpha$ and $u = \sin\alpha$. The relationship between $u$ and $z$ is given by

$$u = \frac{y}{\sqrt{y^2+20}}$$

therefore

$$E_y = \frac{\rho_S}{\pi\varepsilon}\times\frac{1}{2}\arctan\frac{u}{2}\Bigg|_{=-3/\sqrt{29}}^{u=3/\sqrt{29}}$$

$$= \frac{5\times10^{-6}}{\pi\times\frac{1}{36\pi}\times10^{-9}}\times\frac{1}{2}\times2\arctan\frac{\frac{3}{\sqrt{29}}}{2} = 4.8898\times10^4$$

$$\mathbf{E} = E_y\mathbf{a}_y = 4.8898\times10^4\mathbf{a}_y \quad \text{V/m}$$

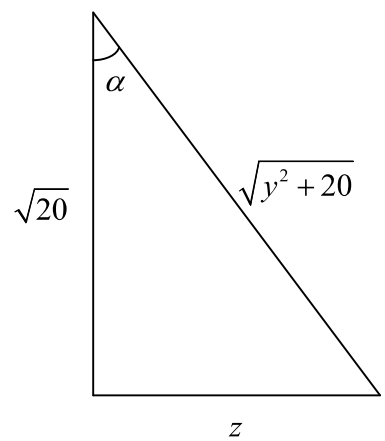


**Figure 4.4** $\boldsymbol{u = \sin\alpha = \frac{y}{\sqrt{y^2+20}}}$.

**MATLAB Solution:**

To write a MATLAB code to solve this problem, we equally divide the surface into many cells each with a length $\Delta y$ and a width $\Delta x$. Each cell has a charge of $\Delta Q = \rho_s \Delta x \Delta y$. When $\Delta x$ and $\Delta y$ are very small, the electric field generated by this cell is very close to that generated by a point charge with a charge $\Delta Q$ located at the center of the cell. Hence the electric field generated by the surface charge at point P is given by

$$\mathbf{E} \approx \sum_{j=1}^{m}\sum_{i=1}^{n} \Delta\mathbf{E}_{j,i} = \sum_{j=1}^{m}\sum_{i=1}^{n} \frac{\rho_s \Delta x \Delta y}{4\pi\varepsilon\,|\mathbf{R}_{j,i}|^3}\mathbf{R}_{j,i}$$

where $\mathbf{R}_{j,i}$ is the vector pointing from the center of a cell to the observation point, as shown in Figure 4.5. The location of the center of a cell is given by $x = -2 + \frac{\Delta x}{2} + \Delta x(i-1)$ , $y = -3 + \frac{\Delta y}{2} + \Delta y(j-1)$ and $z = 0$. The MATLAB code is given in the next page.

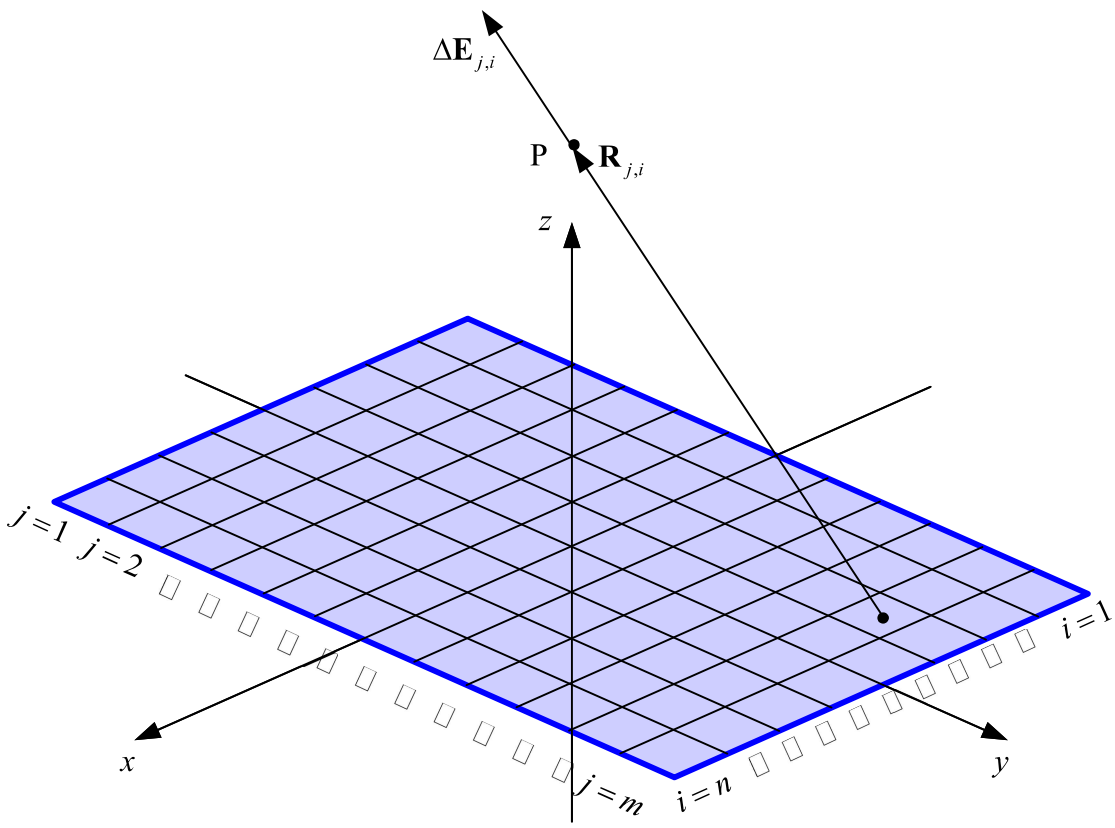


**Figure 4.5 The utilized discretization in the MATLAB code.**

**MATLAB code:**

```
clc; %clear the command line
clear; %remove all previous variables

Epsilono=8.854e-12; %use permittivity of air
D=5e-6; %the surface charge density
P=[0 0 4]; %the position of the observation point
E=zeros(1,3); % initialize E=(0 ,0, 0)

Number_of_x_Steps=100;%initialize discretization in the x direction
Number_of_y_Steps=100;%initialize discretization in %the z direction


x_lower=-2; %the lower boundary of x
x_upper=2; %the upper boundary of x
y_lower=-3; %the lower boundary of y
y_upper=-2; %the upper boundary of y
dx=(x_upper- x_lower)/Number_of_x_Steps; %the x increment or the width of a grid
dy=(y_upper- y_lower)/Number_of_y_Steps; %The y increment or the length of a grid
ds=dx*dy; %the area of a single grid
dQ=D*ds; % the charge on a single grid

for j=1: Number_of_y_Steps
    for i=1: Number_of_x_Steps
        x= x_lower +dx/2+(i-1)*dx; %the x component of the center of a grid
        y= y_lower +dy/2+(j-1)*dy; %the y component of the center of a grid
        R=P-[x y 0];%  vector R is the vector seen from the center of the grid to the
observation point
        RMag=norm(R); % magnitude of vector R
        E=E+(dQ/(4*Epsilono*pi* RMag ^3))*R; % get contribution to the E field
    end
end
```

**Running result:**

```
Command Window
>> E

E =

  1.0e+004 *

   -0.0000    0.0000    4.8833

>>
```

Comparing the MATLAB answer and the analytical answer we see that there is a slight difference. This difference is a result of the finite discretization of the surface *S*.

**Exercise:** Given the surface charge density, $\rho_s$=2.0 $\mu$C/m$^2$, existing in the region $\rho$<1.0 m, $z$ = 0, and zero elsewhere, find **E** at $P$ ( $\rho$ = 0, $z$ = 1.0) and write a MATLAB program to verify your answer.

.

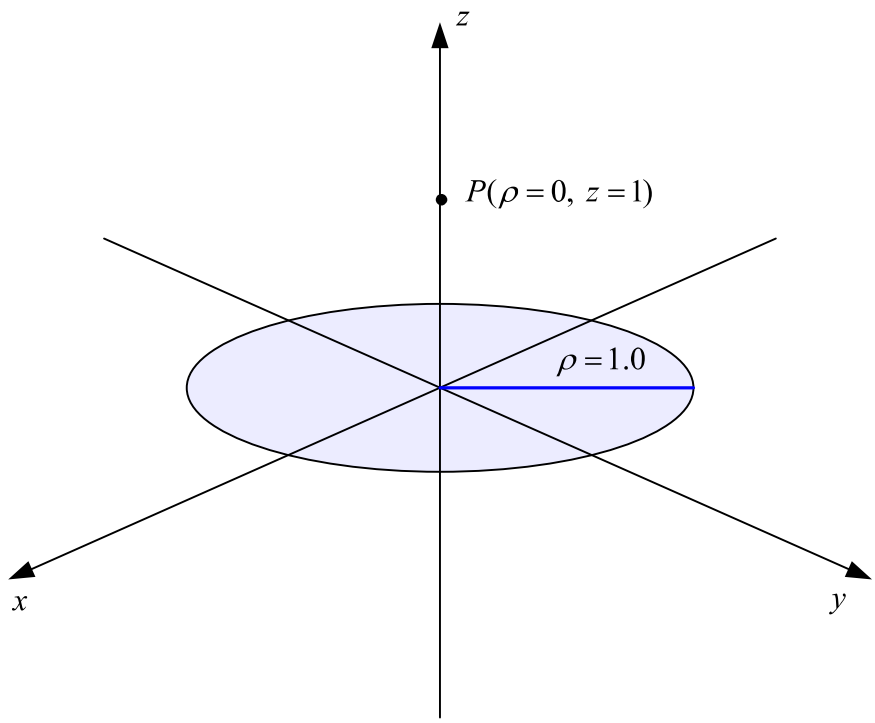


**Figure 4.6 The problem of the exercise of Set 4.**